a

Seismic surveys active on 8 March 2014 and the stations that heard them

10 s survey: Ramform Explorer reported position (Block AD-7 3D survey)
10 s survey, 95% region from timing and paths only
10 s survey, 95% region with 35° bearing from Diego Garcia
10 s survey, 50% region with 35° bearing
7.8 s survey: TGS Huzzas 3D survey area
Huzzas vessel on 8 March (navigation bearings 112.5–112.7°)
Woodside Centaurus 3D survey area (24 Feb–18 Mar; not heard)
Woodside Babylon 3D survey area (21 Mar–14 Apr; not active on 8 March)
Bearings from Diego Garcia South (CMST): 35.2° and 113.3°, shaded = range
Bearings from Diego Garcia South (Stead): 29.8° and 117°

10 s survey position density, timing and paths only (log10, relative to peak)

H08S
MAD
NEAMS (recorder lost)
SSEIR
NCRO
SWAMS
WKER
Scott Reef
Perth Canyon
H01W

Stations (pair of symbols)
heard
possible (p ≈ 0.005–0.05)
not detected, sound path open
sound path blocked by land or shelf
left symbol: 10 s survey
right symbol: 7.8 s survey

Latitude (°N)
Longitude (°E)

b

Northern Bay of Bengal: the 10 s survey

Ramform Explorer, 00:46 UTC

North West Cape: the 7.8 s survey and two others

Centaurus
(1,000–1,200 m water)
solid grey line: 200 m
dashed grey line: 1,000 m
Huzzas
113.3°
Babylon (dotted;
started 21 March)

Note: The map shows each seismic survey that was operating on 8 March 2014 near the sound paths to the stations. The 10 s survey is in the north of the Bay of Bengal (Block AD-7, off Myanmar). The 7.8 s survey is the TGS Huzzas survey off North West Cape, Australia. The Woodside Centaurus survey was also operating, but no station recorded it. The Babylon survey started on 21 March, so it was not operating. Each station has two symbols: the left symbol is for the 10 s survey, the right symbol is for the 7.8 s survey.

Technical note: (a,b) Survey position density on a 0.5° grid (30S–25N, 40–110E, ocean cells, uniform prior); likelihood from SWAMS3−SSEIR delay 577 ± 8 s, NCRO and WKER period-match likelihoods, and GEBCO_2026 path states; red contours add CMST bearing 35.15 ± 1.0°. Ramform Explorer position 19.911973N 91.971170E at 00:46:18 UTC as reported by 370location.org (vessel-to-AD-7 link not independently confirmed). (c) Huzzas polygon from NOPSEMA; vessel band = CMST navigation-derived bearings 112.52–112.73° from H08S inside the polygon; CMST acoustic bearing 113.0–113.6° (0–2.14 h), offset ≈0.6°. Centaurus and Babylon acquisition areas from Woodside EP summary (NOPSEMA A333595, Tables 1-1, 1-3, GDA94); dates from Woodside 2014 Half-Year Report (Centaurus 24 Feb–18 Mar, Babylon 21 Mar–14 Apr). Stead (LA-UR-14-24972) DOAs 29.8° and 117° are single samples. Detection states: 10 s train from Royer (2016) spectrograms and Kadri et al. (2024) Fig. 9; 7.8 s train from period scans (SSEIR 7.938 s p = 0.005, ≈0.1 after multiple-record correction; NCRO p ≈ 0.05). Path blocked = land, or any point shallower than 200 m, on the GEBCO_2026 great-circle path (first 50 km at the source and 20–60 km at the receiver excluded). Not shown: H08N (all paths blocked by the Chagos Bank) and Portland 3274 (141.2E; blocked for both surveys).